Cart-pole model and PID control

Consider a cart of mass $M > 0$ moving along a horizontal track, with a pendulum of mass $m > 0$ attached to it. The pendulum is represented by a point mass at the end of a massless rigid rod of length $\ell > 0$. The pivot is frictionless, and air resistance and friction on the cart are neglected. Let $p(t) \in \mathbb{R}$ denote the horizontal position of the cart, $\theta(t)$ the angle measured from the downward vertical, and $u(t) \in \mathbb{R}$ the horizontal force applied to the cart. A positive angle moves the mass to the right when the pendulum is near the downward position. The objective is to keep the pendulum upright and bring the cart to $p = 0$.

Taking the pivot height as zero, the coordinates of the pendulum mass are

$$x_b = p + \ell \sin\theta, \quad y_b = -\ell \cos\theta. \tag{1}$$

Differentiating gives

$$\dot{x}_b = \dot{p} + \ell\dot{\theta}\cos\theta, \tag{2}$$

$$\dot{y}_b = \ell\dot{\theta}\sin\theta, \tag{3}$$

$$\dot{x}_b^2 + \dot{y}_b^2 = \dot{p}^2 + 2\ell\dot{p}\dot{\theta}\cos\theta + \ell^2\dot{\theta}^2. \tag{4}$$

Consequently, the kinetic energy T and gravitational potential energy V are

$$T = \frac{1}{2}M\dot{p}^2 + \frac{1}{2}m(\dot{x}_b^2 + \dot{y}_b^2) = \frac{1}{2}(M+m)\dot{p}^2 + m\ell\dot{p}\dot{\theta}\cos\theta + \frac{1}{2}m\ell^2\dot{\theta}^2, \tag{5}$$

$$V = mgy_b = -mg\ell\cos\theta, \tag{6}$$

where $g > 0$ is gravitational acceleration. Define the Lagrangian $L = T - V$. The applied force acts on the cart, while no torque is applied directly to the pendulum. Thus, the Euler–Lagrange equations are

$$\frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial L}{\partial \dot{p}} - \frac{\partial L}{\partial p} = u, \quad \frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial L}{\partial \dot{\theta}} - \frac{\partial L}{\partial \theta} = 0. \tag{7}$$

For the first equation, we have

$$\frac{\partial L}{\partial \dot{p}} = (M+m)\dot{p} + m\ell\dot{\theta}\cos\theta, \tag{8}$$

$$\frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial L}{\partial \dot{p}} = (M+m)\ddot{p} + m\ell\ddot{\theta}\cos\theta - m\ell\dot{\theta}^2\sin\theta, \tag{9}$$

$$\frac{\partial L}{\partial p} = 0. \tag{10}$$

For the second equation,

$$\frac{\partial L}{\partial \dot{\theta}} = m\ell\dot{p}\cos\theta + m\ell^2\dot{\theta}, \tag{11}$$

$$\frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial L}{\partial \dot{\theta}} = m\ell\ddot{p}\cos\theta - m\ell\dot{p}\dot{\theta}\sin\theta + m\ell^2\ddot{\theta}, \tag{12}$$

$$\frac{\partial L}{\partial \theta} = -m\ell\dot{p}\dot{\theta}\sin\theta - mg\ell\sin\theta. \tag{13}$$

Hence, the model is

$$(M+m)\ddot{p} + m\ell\cos\theta\,\ddot{\theta} - m\ell\sin\theta\,\dot{\theta}^2 = u, \tag{14}$$

$$m\ell\cos\theta\,\ddot{p} + m\ell^2\ddot{\theta} + mg\ell\sin\theta = 0. \tag{15}$$

With this angle convention, $\theta = 0$ is the downward position and $\theta = \pi$ is the upright position.

For the control, it is more convenient to use an angle equal to zero at the desired position. Define

$$\alpha := \pi - \theta, \quad \dot{\alpha} = -\dot{\theta}, \quad \ddot{\alpha} = -\ddot{\theta}. \tag{16}$$

Then $\sin\theta = \sin\alpha$ and $\cos\theta = -\cos\alpha$. The mass has coordinates $(p + \ell\sin\alpha, \ell\cos\alpha)$, so a small positive α means that the pole leans to the right. Substituting into (15) gives

$$(M+m)\ddot{p} + m\ell\cos\alpha\,\ddot{\alpha} - m\ell\sin\alpha\,\dot{\alpha}^2 = u, \tag{17}$$

$$\ell\ddot{\alpha} = g\sin\alpha - \ddot{p}\cos\alpha. \tag{18}$$

Replacing $\ell\ddot{\alpha}$ in the first equation yields

$$(M + m - m\cos^2\alpha)\ddot{p} = u + m\ell\dot{\alpha}^2\sin\alpha - mg\sin\alpha\cos\alpha. \tag{19}$$

Since $M + m - m\cos^2\alpha = M + m\sin^2\alpha > 0$, the accelerations are well defined,

$$\ddot{p} = \frac{u + m\sin\alpha(\ell\dot{\alpha}^2 - g\cos\alpha)}{M + m\sin^2\alpha}, \tag{20}$$

$$\ddot{\alpha} = \frac{g\sin\alpha - \ddot{p}\cos\alpha}{\ell}. \tag{21}$$

Let $\dot{p} = v$ and $\dot{\alpha} = \omega$, these equations define the nonlinear state model for $X = (p, v, \alpha, \omega)^\top \in \mathbb{R}^4$. The desired equilibrium is $X = 0$, with $u = 0$.

We linearize the system around the steady-state operating point $(x_{\text{ss}}, u_{\text{ss}}) = (0, 0)$, where $x = (p, \dot{p}, \alpha, \dot{\alpha})^\top$. Writing the nonlinear system as $\dot{x} = f(x, u)$, the first-order Taylor series expansion gives

$$\begin{aligned}\frac{d}{dt}(x - x_{\text{ss}}) &\simeq f(x_{\text{ss}}, u_{\text{ss}}) + \left.\frac{\partial f}{\partial x}\right|_{(x_{\text{ss}}, u_{\text{ss}})}(x - x_{\text{ss}}) + \left.\frac{\partial f}{\partial u}\right|_{(x_{\text{ss}}, u_{\text{ss}})}(u - u_{\text{ss}}) \\ &= A(x - x_{\text{ss}}) + B(u - u_{\text{ss}}),\end{aligned} \tag{22}$$

with $f(x_{\text{ss}}, u_{\text{ss}}) = 0$ and

$$A = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & -\dfrac{mg}{M} & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & \dfrac{(M+m)g}{M\ell} & 0 \end{pmatrix}, \quad B = \begin{pmatrix} 0 \\ \dfrac{1}{M} \\ 0 \\ -\dfrac{1}{M\ell} \end{pmatrix}. \tag{23}$$

Thus, the linearized equations are

$$\ddot{p} = \frac{u - mg\alpha}{M}, \quad \ddot{\alpha} = \frac{(M+m)g}{M\ell}\alpha - \frac{u}{M\ell}. \tag{24}$$

Without control ($u = 0$), the linearized angle equation has characteristic roots $\lambda = \pm\sqrt{(M+m)g/(M\ell)}$. The positive root implies that the upright equilibrium is unstable.

We first consider an angle PD control $u = K\alpha + D\dot{\alpha}$. Substituting into (24) gives

$$\ddot{\alpha} + \frac{D}{M\ell}\dot{\alpha} + \frac{K - (M+m)g}{M\ell}\alpha = 0. \tag{25}$$

Thus $D > 0$ and $K > (M+m)g$ stabilize the angle in the linear model. The proportional term counteracts the instability, while the derivative term damps the motion. Without the derivative term, i.e., $D = 0$, the angle oscillates with constant amplitude. However, angle control alone does not require $p(t) \to 0$, so the cart may drift.

To also control the cart, define its error and integral by

$$e_p = -p, \quad I_p(t) = \int_0^t e_p(\tau)\,\mathrm{d}\tau, \quad \alpha_d = \kappa_p e_p + \kappa_i I_p + \kappa_d \dot{e}_p. \tag{26}$$

This is a position PID whose output is a desired lean angle, or the angle needed to adjust position. Let $e_\alpha = \alpha_d - \alpha$. The inner angle loop uses

$$u = -Ke_\alpha + D\dot{\alpha} = K(\alpha - \alpha_d) + D\dot{\alpha}. \tag{27}$$

The sign follows from the negative coefficient of u in the angle equation. The derivative acts on the measured angle, rather than on the changing reference. For example, suppose the outer loop suddenly changes the desired angle α_d, the pendulum has not moved yet, so $\dot{\alpha} = 0$, then, the proportional term responds to the new target, while the derivative term responds only when the pendulum actually moves. Both loops use the observed velocities directly in the simulation.

Let $z(t) = \displaystyle\int_0^t p(\tau)\,\mathrm{d}\tau = -I_p(t)$, so $\dot{z} = p$ and $z(0) = 0$. Combining (26) and (27) gives

$$u = K\alpha + D\dot{\alpha} + k_p p + k_d \dot{p} + k_i z, \quad \dot{z} = p, \tag{28}$$

$$k_p = K\kappa_p, \quad k_d = K\kappa_d, \quad k_i = K\kappa_i. \tag{29}$$

The position integral corrects a persistent cart offset, for example under a small constant horizontal disturbance. The angle PD loop keeps the pole close to upright while the position PID changes the desired lean to move the cart back.

Choosing each gain independently does not guarantee stability of the coupled system. We therefore compute the characteristic polynomial of the full linearized model, including z.

Write

$$F = \begin{pmatrix} k_p & k_d & K & D \end{pmatrix}, \quad C_p = \begin{pmatrix} 1 & 0 & 0 & 0 \end{pmatrix}. \tag{30}$$

The controller (28) gives $u = Fx + k_i z$ and $\dot{z} = C_p x$. Substituting the control into the linearization (22) and adding the integral state z yields

$$\dot{\xi} = A_c \xi, \quad \xi = \begin{pmatrix} x \\ z \end{pmatrix} = \begin{pmatrix} p & \dot{p} & \alpha & \dot{\alpha} & z \end{pmatrix}^\top, \quad A_c = \begin{pmatrix} A + BF & Bk_i \\ C_p & 0 \end{pmatrix}. \tag{31}$$

Explicitly,

$$A_c = \begin{pmatrix} 0 & 1 & 0 & 0 & 0 \\ \dfrac{k_p}{M} & \dfrac{k_d}{M} & \dfrac{K - mg}{M} & \dfrac{D}{M} & \dfrac{k_i}{M} \\ 0 & 0 & 0 & 1 & 0 \\ -\dfrac{k_p}{M\ell} & -\dfrac{k_d}{M\ell} & \dfrac{(M+m)g - K}{M\ell} & -\dfrac{D}{M\ell} & -\dfrac{k_i}{M\ell} \\ 1 & 0 & 0 & 0 & 0 \end{pmatrix}. \tag{32}$$

To have the closed-loop system (31) be asymptotically (exponentially) stable, we want the matrix A_c to be Hurwitz, i.e., eigenvalues of A_c be on the left hand side of the complex plane.

The elimination below provides another way to compute this characteristic equation. From (24),

$$M\ddot{p} = u - mg\alpha, \quad M\ell\ddot{\alpha} = (M+m)g\alpha - u. \tag{33}$$

For a mode proportional to e^{st}, write the amplitudes of p and α as P and A_α. The linear mechanical equations give

$$(M+m)s^2 P + m\ell s^2 A_\alpha = U, \tag{34}$$

$$s^2 P + (\ell s^2 - g)A_\alpha = 0. \tag{35}$$

For $s \neq 0$, the integral-state amplitude is $Z = P/s$, so the control gives

$$U = (K + Ds)A_\alpha + \left(k_d s + k_p + \frac{k_i}{s}\right) P. \tag{36}$$

For $\ell s^2 \neq g$, (35) gives

$$A_\alpha = -\frac{s^2}{\ell s^2 - g} P. \tag{37}$$

Substitute (36) and (37) into (34), then move everything to the left

$$\left[(M+m)s^2 - k_d s - k_p - \frac{k_i}{s} - \frac{(m\ell s^2 - K - Ds)s^2}{\ell s^2 - g}\right] P = 0. \tag{38}$$

Multiplying by $s(\ell s^2 - g)$ gives $\chi(s)P = 0$, where we define the polynomial

$$\chi(s) := s\big(M\ell s^4 - (M+m)gs^2\big) + (K + Ds)s^3 - (k_d s^2 + k_p s + k_i)(\ell s^2 - g). \tag{39}$$

For a mode with $P \neq 0$, this requires $\chi(s) = 0$. More generally, a direct determinant calculation using (32) gives the polynomial identity

$$\det(sI_5 - A_c) = \frac{\chi(s)}{M\ell}. \tag{40}$$

This identity is valid for every s, including the values excluded by the divisions in the elimination. Consequently, the roots of χ are exactly the eigenvalues of A_c. The factor $M\ell$ appears because χ has leading term $M\ell s^5$ Since $M\ell > 0$, dividing by it does not change the roots. Expanding gives

$$\begin{aligned}\chi(s) = M\ell s^5 + (D - \ell k_d)s^4 + \big(K - (M+m)g - \ell k_p\big)s^3 \\ + (gk_d - \ell k_i)s^2 + gk_p s + gk_i.\end{aligned} \tag{41}$$

Choose positive numbers $r_1, \ldots, r_5$ and require the roots to be $-r_1, \ldots, -r_5$, so that the linearized closed loop is exponentially stable and Lyapunov's indirect method applies to the nonlinear equilibrium. Write

$$\prod_{j=1}^{5}(s + r_j) = s^5 + a_1 s^4 + a_2 s^3 + a_3 s^2 + a_4 s + a_5. \tag{42}$$

Matching (41) with $M\ell \prod_{j=1}^{5}(s + r_j)$ yields

$$k_i = \frac{M\ell a_5}{g}, \tag{43}$$

$$k_p = \frac{M\ell a_4}{g}, \tag{44}$$

$$k_d = \frac{M\ell a_3 + \ell k_i}{g}, \tag{45}$$

$$D = M\ell a_1 + \ell k_d, \tag{46}$$

$$K = M\ell a_2 + (M+m)g + \ell k_p, \tag{47}$$

where

$$a_1 = \sum_{i=1}^{5} r_i, \tag{48}$$

$$a_2 = \sum_{1 \leq i < j \leq 5} r_i r_j, \tag{49}$$

$$a_3 = \sum_{1 \leq i < j < k \leq 5} r_i r_j r_k, \tag{50}$$

$$a_4 = \sum_{1 \leq i < j < k < l \leq 5} r_i r_j r_k r_l, \tag{51}$$

$$a_5 = r_1 r_2 r_3 r_4 r_5. \tag{52}$$

For the numerical simulation, choose

$$M = 1 \text{ kg}, \quad m = 0.1 \text{ kg}, \quad \ell = 0.5 \text{ m}, \quad g = 9.81 \text{ m s}^{-2}, \tag{53}$$

$$(r_1, r_2, r_3, r_4, r_5) = (1, 1.2, 2, 3, 4) \text{ s}^{-1}. \tag{54}$$

The desired polynomial is

$$(s+1)(s+1.2)(s+2)(s+3)(s+4) = s^5 + 11.2s^4 + 47s^3 + 92s^2 + 84s + 28.8. \tag{55}$$

Consequently, (43)–(47) gives

$$K \simeq 36.431673, \quad D \simeq 7.981954, \quad k_p \simeq 4.281346, \quad k_d \simeq 4.763909, \quad k_i \simeq 1.467890. \tag{56}$$

The corresponding position PID gains are

$$\kappa_p = \frac{k_p}{K} \simeq 0.117517, \quad \kappa_i = \frac{k_i}{K} \simeq 0.040292, \quad \kappa_d = \frac{k_d}{K} \simeq 0.130763. \tag{57}$$

All the chosen roots have negative real part, so the five-state linearized closed loop converges exponentially to zero. With the unsaturated control, and states define from (31), dynamics of the nonlinear original problem is smooth near zero and can be written as

$$\dot{\xi} = A_c\xi + r(\xi), \quad \frac{\|r(\xi)\|}{\|\xi\|} \longrightarrow 0 \quad \text{as } \xi \to 0, \tag{58}$$

where r is the nonlinear reminder. Since all its eigenvalues have negative real part, the matrix

$$P = \int_0^\infty e^{A_c^\top t} e^{A_c t} \, \mathrm{d}t \succ 0 \tag{59}$$

is finite and satisfies $A_c^\top P + PA_c = -I$. Define $W(\xi) := \xi^\top P \xi > 0$, we have

$$\dot{W} = \dot{\xi}^\top P\xi + \xi^\top P\dot{\xi}. \tag{60}$$

$$= (A_c\xi + r(\xi))^\top P\xi + \xi^\top P(A_c\xi + r(\xi)) \tag{61}$$

$$= \xi^\top (A_c^\top P + PA_c)\xi + r(\xi)^\top P\xi + \xi^\top Pr(\xi) \tag{62}$$

$$= -\|\xi\|^2 + 2\xi^\top Pr(\xi) \le -\|\xi\|^2 + 2\|P\|\|\xi\|\|r(\xi)\|. \tag{63}$$

Since

$$\frac{\|r(\xi)\|}{\|\xi\|} \longrightarrow 0 \quad \text{as } \xi \to 0. \tag{64}$$

Then, there exists $\delta > 0$ such that

$$0 < \|\xi\| < \delta \quad \Longrightarrow \quad \frac{\|r(\xi)\|}{\|\xi\|} \le \frac{1}{4\|P\|}. \tag{65}$$

Hence, for sufficiently small ξ, the last term is at most $\frac{1}{2}\|\xi\|^2$. Therefore $\dot{W} \le -\frac{1}{2}\|\xi\|^2$, which gives local exponential stability of the nonlinear equilibrium. In particular,

$$p(t) \to 0, \quad \dot{p}(t) \to 0, \quad \alpha(t) \to 0, \quad \dot{\alpha}(t) \to 0. \tag{66}$$

This is a local result. It does not assert that the PID control can raise the pendulum from its downward position.

The simulation limits the actuator to $u_{\max} = 10$ N. At sample time $t_k = kh$, with $h = 0.01$ s, it computes

$$u_k^{\text{raw}} = K\alpha_k + D\omega_k + k_p p_k + k_d v_k + k_i z_k, \tag{67}$$

$$u_k = \text{sat}_{[-u_{\max}, u_{\max}]}(u_k^{\text{raw}}). \tag{68}$$

To avoid accumulating an integral that would worsen actuator saturation, it uses

$$z_{k+1} = \begin{cases} z_k, & (u_k^{\text{raw}} - u_k) k_i p_k > 0, \\ z_k + h p_k, & \text{otherwise.} \end{cases} \tag{69}$$

The choice for this particular integral can be seen by fixing all other control terms as a and compare $a + k_i z_{k+1}$ with $u_k^{\text{raw}} = a + k_i z_k$. The proposed integral update changes the force by $h k_i p_k$.

The accompanying Python program uses a custom Gymnasium environment with observation (p, v, α, ω) and continuous action u. The exact nonlinear mechanical equations (21) are integrated with fourth-order Runge–Kutta, holding u_k fixed over $[t_k, t_{k+1})$. Starting from

$$p(0) = 0.25 \text{ m}, \quad \dot{p}(0) = 0, \quad \alpha(0) = 8^\circ, \quad \dot{\alpha}(0) = 0, \quad z(0) = 0, \tag{70}$$

the 15-second run gives approximately

$$p(15) = -5.37 \times 10^{-6} \text{ m}, \qquad \alpha(15) = -5.93 \times 10^{-7} \text{ rad}, \tag{71}$$

$$\max_k |p_k| = 0.45013 \text{ m}, \qquad \max_k |u_k| = 6.15716 \text{ N}. \tag{72}$$

The pole returns to upright while the cart returns to the origin.

To apply the control to the built-in Gymnasium environment, the subsequent changes are needed. First, Gymnasium's equations include a 4/3 inertia factor, and its length parameter represents half the pole's full length. This corresponds to a uniform rod of length 2ℓ, whose centre of mass is at distance ℓ. For that rod, the moment of inertia about its centre is

$$I_{\text{cm}} = \frac{1}{12} m (2\ell)^2 = \frac{1}{3} m \ell^2. \tag{73}$$

Consequently, the kinetic energy becomes

$$T = \frac{1}{2}(M+m)\dot{p}^2 + m\ell \dot{p}\dot{\alpha}\cos\alpha + \frac{1}{2}(m\ell^2 + I_{\text{cm}})\dot{\alpha}^2 \tag{74}$$

$$= \frac{1}{2}(M+m)\dot{p}^2 + m\ell \dot{p}\dot{\alpha}\cos\alpha + \frac{2}{3} m\ell^2 \dot{\alpha}^2. \tag{75}$$

The potential energy remains

$$V = mg\ell\cos\alpha. \tag{76}$$

Applying Euler-Lagrange gives

$$(M+m)\ddot{p} + m\ell\cos\alpha\,\ddot{\alpha} - m\ell\sin\alpha\,\dot{\alpha}^2 = u, \tag{77}$$

$$\frac{4}{3}\ell\ddot{\alpha} = g\sin\alpha - \ddot{p}\cos\alpha. \tag{78}$$

Solving for the accelerations gives

$$\ddot{p} = \frac{\frac{4}{3}u + m\sin\alpha\left(\frac{4}{3}\ell\dot{\alpha}^2 - g\cos\alpha\right)}{\frac{4}{3}(M+m) - m\cos^2\alpha}, \tag{79}$$

$$\ddot{\alpha} = \frac{g\sin\alpha - \ddot{p}\cos\alpha}{\frac{4}{3}\ell}. \tag{80}$$

The new linearization near the operating steady state is

$$\ddot{p} = \frac{\frac{4}{3}u - mg\alpha}{\frac{4}{3}(M+m) - m}, \qquad \ddot{\alpha} = \frac{(M+m)g\alpha - u}{\ell\left(\frac{4}{3}(M+m) - m\right)}. \tag{81}$$

The characteristic polynomial is then

$$\chi(s) = \left(\frac{4}{3}(M+m) - m\right)\ell s^5 + \left(D - \frac{4}{3}\ell k_d\right)s^4 + \left(K - (M+m)g - \frac{4}{3}\ell k_p\right)s^3 \tag{82}$$

$$+ \left(gk_d - \frac{4}{3}\ell k_i\right)s^2 + gk_p s + gk_i. \tag{83}$$

Coefficient matching now gives

$$k_i = \frac{\left(\frac{4}{3}(M+m) - m\right)\ell a_5}{g}, \tag{84}$$

$$k_p = \frac{\left(\frac{4}{3}(M+m) - m\right)\ell a_4}{g}, \tag{85}$$

$$k_d = \frac{\left(\frac{4}{3}(M+m) - m\right)\ell a_3 + \frac{4}{3}\ell k_i}{g}, \tag{86}$$

$$D = \left(\frac{4}{3}(M+m) - m\right)\ell a_1 + \frac{4}{3}\ell k_d, \tag{87}$$

$$K = \left(\frac{4}{3}(M+m) - m\right)\ell a_2 + (M+m)g + \frac{4}{3}\ell k_p. \tag{88}$$

Finally, the built-in CartPole applies

$$u = \begin{cases} -10\,\mathrm{N}, & \text{action } 0, \\ +10\,\mathrm{N}, & \text{action } 1. \end{cases} \tag{89}$$

There is no changing force magnitude. A simple adaptation is

$$\text{action}_k = \begin{cases} 1, & u_{\mathrm{PID},k} \geq 0, \\ 0, & u_{\mathrm{PID},k} < 0. \end{cases} \tag{90}$$